

```
1 "D:\HigherStudies\metu\MS Electrical and Electronics Engineering\ee574\Project
Material\ownCode\build_from_scratch\.venv\Scripts\python.exe" "D:\HigherStudies\metu\MS Electrical and Electronics
Engineering\ee574\Project Material\ownCode\metu_state_estimator_pipeline1bdd.py"
```

EE574 - AC WLS State Estimator

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Pipeline:

1. Parse CDF network file
2. Parse measurement file (8 blocks)
3. Build Y-bus
4. Identify PMU buses from Block 2 (V_ang)
5. Assemble measurement vector z and weight matrix W
6. Observability analysis (SVD rank of H)
7. AC-WLS estimation (Newton iteration)
8. Bad data detection & identification (LNR)

Please write the name of CDF file with extension: example2cdf.dat

Please write the name of compatible measurement file with extension: example2measure.dat

SECTION 1 - PARSING IEEE CDF NETWORK FILE

[CDF] File: example2cdf.dat

mode is None

example2cdf 100.0

100.0

[CDF] Base MVA = 100.0

[CDF] Line 2: → BUS DATA block

[CDF] Line 17: end-of-block (-999)

[CDF] Line 18: → BRANCH DATA block

[CDF] Line 39: end-of-block (-999)

[CDF] Line 45: end-of-block (-999)

[CDF] ✓ Buses parsed : 14

[CDF] ✓ Branches parsed : 20

Bus	Name	Type	V pu	θ°	PLpu	QLpu	PGpu	QGpu
1		Slack	1.06	0	0	0	2.3228	-0.168
2		PV	1.045	-4.98	0.217	0.127	0.4	0.423
3		PV	1.01	-12.72	0.942	0.19	0	0.233
4		PQ	1.0186	-10.32	0.478	-0.039	0	0
5		PQ	1.0203	-8.78	0.076	0.016	0	0
6		PV	1.07	-14.22	0.112	0.075	0	0.122

101		8	P_inj	pu		12	-		-0.061		0.001		0
102	+-----+												
103		9	P_inj	pu		13	-		-0.135		0.001		0
104	+-----+												
105		10	P_inj	pu		6	-		-0.112		0.001		0
106	+-----+												
107		11	P_inj	pu		9	-		-0.295		0.001		0
108	+-----+												
109		12	P_inj	pu		4	-		-0.478		0.001		0
110	+-----+												
111		13	P_inj	pu		5	-		-0.076		0.001		0
112	+-----+												
113		14	P_inj	pu		11	-		-0.035		0.001		0
114	+-----+												
115		15	Q_inj	pu		14	-		-0.05		0.001		0
116	+-----+												
117		16	Q_inj	pu		10	-		-0.058		0.001		0
118	+-----+												
119		17	Q_inj	pu		7	-		0		0.001		0
120	+-----+												
121		18	Q_inj	pu		8	-		0.17357		0.001		0
122	+-----+												
123		19	Q_inj	pu		3	-		0.04393		0.001		0
124	+-----+												
125		20	Q_inj	pu		2	-		0.29695		0.001		0
126	+-----+												
127		21	Q_inj	pu		1	-		-0.16888		0.001		0
128	+-----+												
129		22	Q_inj	pu		12	-		-0.016		0.001		0
130	+-----+												
131		23	Q_inj	pu		13	-		-0.058		0.001		0
132	+-----+												
133		24	Q_inj	pu		6	-		0.04718		0.001		0
134	+-----+												
135		25	Q_inj	pu		9	-		-0.166		0.001		0
136	+-----+												
137		26	Q_inj	pu		4	-		0.039		0.001		0
138	+-----+												
139		27	Q_inj	pu		5	-		-0.016		0.001		0
140	+-----+												
141		28	Q_inj	pu		11	-		-0.018		0.001		0
142	+-----+												
143		29	P_flow	pu		12	13		0.0161		0.001		0
144	+-----+												
145		30	P_flow	pu		5	4		0.61735		0.001		0
146	+-----+												
147		31	P_flow	pu		5	6		0.44061		0.001		0
148	+-----+												
149		32	P_flow	pu		13	14		0.05634		0.001		0
150	+-----+												
151		33	P flow	pu		2	3		0.73186		0.001		0

152											
153	34	P_flow	pu		6 13		0.17736		0.001		0
154											
155	35	P_flow	pu		1 2		1.56828		0.001		0
156											
157	36	P_flow	pu		3 4		-0.23329		0.001		0
158											
159	37	P_flow	pu		7 8		0		0.001		0
160											
161	38	P_flow	pu		4 7		0.28086		0.001		0
162											
163	39	P_flow	pu		12 6		-0.07708		0.001		0
164											
165	40	P_flow	pu		14 9		-0.09321		0.001		0
166											
167	41	P_flow	pu		10 9		-0.05226		0.001		0
168											
169	42	P_flow	pu		11 10		0.03789		0.001		0
170											
171	43	P_flow	pu		6 11		0.07339		0.001		0
172											
173	44	P_flow	pu		2 5		0.41513		0.001		0
174											
175	45	P_flow	pu		1 5		0.75552		0.001		0
176											
177	46	P_flow	pu		2 4		0.56139		0.001		0
178											
179	47	P_flow	pu		9 7		-0.28085		0.001		0
180											
181	48	P_flow	pu		9 4		-0.16089		0.001		0
182											
183	49	Q_flow	pu		12 13		0.00743		0.001		0
184											
185	50	Q_flow	pu		5 4		-0.15366		0.001		0
186											
187	51	Q_flow	pu		5 6		0.12824		0.001		0
188											
189	52	Q_flow	pu		13 14		0.01694		0.001		0
190											
191	53	Q_flow	pu		2 3		0.03565		0.001		0
192											
193	54	Q_flow	pu		6 13		0.07161		0.001		0
194											
195	55	Q_flow	pu		1 2		-0.20391		0.001		0
196											
197	56	Q_flow	pu		3 4		0.02809		0.001		0
198											
199	57	Q_flow	pu		7 8		-0.1691		0.001		0
200											
201	58	Q_flow	pu		4 7		-0.09422		0.001		0
202											

203		59		Q_flow		pu				12		6		-0.02338		0.001		0		
204	+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+																			
205		60		Q_flow		pu				14		9		-0.03417		0.001		0		
206	+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+																			
207		61		Q_flow		pu				10		9		-0.04272		0.001		0		
208	+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+																			
209		62		Q_flow		pu				11		10		0.01564		0.001		0		
210	+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+																			
211		63		Q_flow		pu				6		11		0.03465		0.001		0		
212	+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+																			
213		64		Q_flow		pu				2		5		0.00762		0.001		0		
214	+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+																			
215		65		Q_flow		pu				1		5		0.03504		0.001		0		
216	+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+																			
217		66		Q_flow		pu				2		4		-0.02287		0.001		0		
218	+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+																			
219		67		Q_flow		pu				9		7		-0.04993		0.001		0		
220	+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+																			
221		68		Q_flow		pu				9		4		0.01625		0.001		0		
222	+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+																			

223

224 =====

225 SECTION 3 - BUILDING Y-BUS ADMITTANCE MATRIX

226 =====

227 [YBUS] Matrix size : 14 × 14

228

229 [MAIN] n_buses=14, n_branches=20, n_states=2n-1=27

230 [MAIN] File measurements available: 69

231

232 [MAIN] PMU buses (from Block 2 V_ang records) : []

233 [MAIN] PMU sigma (from file) : 0.001

234 [MAIN] WARNING: No V_ang measurements in file - no PMU buses will be integrated.

235

236 =====

237 SECTION 5 - ASSEMBLING MEASUREMENT VECTOR

238 =====

239 [Z] File measurements loaded : 69

240 [Z] File measurements skipped: 0

241 [Z] File measurements excluded (bad data): 0

242

243 [Z] — Final measurement vector —————

244 [Z] Total measurements m : 69

245 [Z] State dimension 2n-1 : 27

246 [Z] Redundancy m/(2n-1) : 2.556 (>1 required)

247 [Z] Weight 1/σ² range : 1000000.0 ... 1000000.0

248 [Z] Type breakdown:

249 P_flow : 20 σ=0.0010 w=1/σ²=1000000.0

250 P_inj : 14 σ=0.0010 w=1/σ²=1000000.0

251 Q_flow : 20 σ=0.0010 w=1/σ²=1000000.0

252 Q_inj : 14 σ=0.0010 w=1/σ²=1000000.0

253 V_mag : 1 σ=0.0010 w=1/σ²=1000000.0

```

254
255 =====
256 SECTION 8 - OBSERVABILITY ANALYSIS
257 =====
258 [OBS] H shape      : 69 rows × 27 cols
259 [OBS] Required rank : 27 (= 2n-1, n = 14)
260 [OBS] Measurements m : 69
261 [OBS] Redundancy    : 42 extra measurements beyond minimum
262 [OBS] Computed rank  : 27
263 [OBS] Rank deficiency : 0
264
265 [OBS] ✓ System is FULLY OBSERVABLE (rank = 27 = required 27)
266
267 =====
268 SECTION 10 - ITERATIVE BAD DATA REMOVAL LOOP
269 =====
270 [LOOP] Starting iterative WLS + bad-data removal.
271 [LOOP] Rule: remove one measurement per pass (largest r_N),
272 [LOOP]      stop if removal makes system unobservable.
273
274
275 [LOOP] ===== Pass 1 =====
276 [LOOP] Active measurements: 69
277 [LOOP] Excluded so far    : []
278
279 =====
280 SECTION 9 - AC-WLS STATE ESTIMATION (Newton iteration)
281 =====
282 [WLS] Flat-start initialisation: |V|=1.0 pu, θ=0.0 rad
283 [WLS] n_buses      : 14
284 [WLS] n_states (2n-1) : 27 (13 angle states + 14 voltage states)
285 [WLS] Measurements m : 69
286 [WLS] Redundancy    : 69/27 = 2.556
287 [WLS] Convergence tol : 1e-06
288 [WLS] Max iterations : 50
289
290
291 [WLS] ✓ Converged in 6 iterations (max|Δx| = 1.60e-07 < tol = 1e-06)
292 [WLS] Final objective J(x̂) = 15370.928637
293 [WLS] Final □r□□ = 0.123980
294
295 Iteration summary:
296 +-----+-----+-----+-----+-----+
297 | Iter | □r□ | rank (Gm) | cond (Gm) | max|Δx| |
298 +-----+-----+-----+-----+-----+
299 | 1 | 3.49815 | 27 | 61140 | 0.306277 |
300 +-----+-----+-----+-----+-----+
301 | 2 | 0.271647 | 27 | 54360 | 0.0709989 |
302 +-----+-----+-----+-----+-----+
303 | 3 | 0.125227 | 27 | 56430 | 0.0025138 |
304 +-----+-----+-----+-----+-----+

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```
305 |      4 | 0.12398 |      27 |      56220 | 9.9744e-05 |
306 +-----+-----+-----+-----+-----+-----+
307 |      5 | 0.12398 |      27 |      56240 | 3.86869e-06 |
308 +-----+-----+-----+-----+-----+-----+
309 |      6 | 0.12398 |      27 |      56240 | 1.60367e-07 |
310 +-----+-----+-----+-----+-----+-----+
```

```
311
312 Estimated State vs CDF Load-Flow Ground Truth:
```

```
313 +-----+-----+-----+-----+-----+-----+-----+-----+
314 | Bus | Name | |V|_est | |V|_true | Δ|V| | θ_est° | θ_true° | Δθ° |
315 +=====+=====+=====+=====+=====+=====+=====+=====+
316 | 1 | | 1.04331 | 1.06 | -0.01669 | 0 | 0 | 0 |
317 +-----+-----+-----+-----+-----+-----+-----+-----+
318 | 2 | | 1.02807 | 1.045 | -0.01693 | -5.139 | -4.98 | -0.159 |
319 +-----+-----+-----+-----+-----+-----+-----+-----+
320 | 3 | | 0.9929 | 1.01 | -0.0171 | -13.138 | -12.72 | -0.418 |
321 +-----+-----+-----+-----+-----+-----+-----+-----+
322 | 4 | | 1.00089 | 1.0186 | -0.01771 | -10.645 | -10.32 | -0.325 |
323 +-----+-----+-----+-----+-----+-----+-----+-----+
324 | 5 | | 1.00269 | 1.0203 | -0.01761 | -9.054 | -8.78 | -0.274 |
325 +-----+-----+-----+-----+-----+-----+-----+-----+
326 | 6 | | 1.04967 | 1.07 | -0.02033 | -14.672 | -14.22 | -0.452 |
327 +-----+-----+-----+-----+-----+-----+-----+-----+
328 | 7 | | 1.04116 | 1.062 | -0.02084 | -13.78 | -13.37 | -0.41 |
329 +-----+-----+-----+-----+-----+-----+-----+-----+
330 | 8 | | 1.06955 | 1.09 | -0.02045 | -13.776 | -13.37 | -0.406 |
331 +-----+-----+-----+-----+-----+-----+-----+-----+
332 | 9 | | 1.03425 | 1.0563 | -0.02205 | -15.418 | -14.95 | -0.468 |
333 +-----+-----+-----+-----+-----+-----+-----+-----+
334 | 10 | | 1.02907 | 1.0513 | -0.02223 | -15.576 | -15.1 | -0.476 |
335 +-----+-----+-----+-----+-----+-----+-----+-----+
336 | 11 | | 1.03564 | 1.0571 | -0.02146 | -15.256 | -14.79 | -0.466 |
337 +-----+-----+-----+-----+-----+-----+-----+-----+
338 | 12 | | 1.03473 | 1.0552 | -0.02047 | -15.556 | -15.08 | -0.476 |
339 +-----+-----+-----+-----+-----+-----+-----+-----+
340 | 13 | | 1.02959 | 1.0505 | -0.02091 | -15.638 | -15.16 | -0.478 |
341 +-----+-----+-----+-----+-----+-----+-----+-----+
342 | 14 | | 1.01342 | 1.0358 | -0.02238 | -16.544 | -16.04 | -0.504 |
343 +-----+-----+-----+-----+-----+-----+-----+-----+
```

```
344
345 =====
346 SECTION 10 - BAD DATA ANALYSIS [Pass 1]
347 =====
```

```
348 [BDD] m=69, n_states=27, dof=42
349 [BDD] Critical measurements (Ωii≈0, rN forced to 0): 0
350 [BDD] rN stats: min=0.1593 max=123.9777 mean=5.8486
351 [BDD] X BAD DATA DETECTED
```

```
352
353 [BDD] — LNR Test (threshold=3.0) — Top 10 —————
```

```
354 +-----+-----+-----+-----+-----+-----+-----+-----+
355 | Rank | Idx | OrigIdx | Src | Type | Bus_i | Bus_j | z | h(x^ ) | r | r_N | Flag |
```



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407 [LOOP] Tentatively removing measurement orig_idx=68 (r_N=123.9777) ...
408
409 =====
410 SECTION 5 - ASSEMBLING MEASUREMENT VECTOR
411 =====
412 [Z] Excluded measurement original-indices: [68]
413 [Z] SKIP (bad-data-excluded, orig_idx=68): Q_flow bus_i=9 bus_j=4
414 [Z] File measurements loaded : 68
415 [Z] File measurements skipped: 0
416 [Z] File measurements excluded (bad data): 1
417
418 [Z] — Final measurement vector —————
419 [Z] Total measurements m : 68
420 [Z] State dimension 2n-1 : 27
421 [Z] Redundancy m/(2n-1) : 2.519 (>1 required)
422 [Z] Weight 1/σ² range : 1000000.0 ... 1000000.0
423 [Z] Type breakdown:
424     P_flow : 20 σ=0.0010 w=1/σ²=1000000.0
425     P_inj : 14 σ=0.0010 w=1/σ²=1000000.0
426     Q_flow : 19 σ=0.0010 w=1/σ²=1000000.0
427     Q_inj : 14 σ=0.0010 w=1/σ²=1000000.0
428     V_mag : 1 σ=0.0010 w=1/σ²=1000000.0
429
430 [LOOP] Checking observability after removing orig_idx=68 ...
431
432 =====
433 SECTION 8 - OBSERVABILITY ANALYSIS
434 =====
435 [OBS] H shape : 68 rows × 27 cols
436 [OBS] Required rank : 27 (= 2n-1, n = 14)
437 [OBS] Measurements m : 68
438 [OBS] Redundancy : 41 extra measurements beyond minimum
439 [OBS] Computed rank : 27
440 [OBS] Rank deficiency : 0
441
442 [OBS] ✓ System is FULLY OBSERVABLE (rank = 27 = required 27)
443
444 [LOOP] ✓ System remains OBSERVABLE after removing orig_idx=68.
445 [LOOP] Measurement permanently removed. Total excluded: [68]
446
447 [LOOP] ===== Pass 2 =====
448 [LOOP] Active measurements: 68
449 [LOOP] Excluded so far : [68]
450
451 =====
452 SECTION 9 - AC-WLS STATE ESTIMATION (Newton iteration)
453 =====
454 [WLS] Flat-start initialisation: |V|=1.0 pu, θ=0.0 rad
455 [WLS] n_buses : 14
456 [WLS] n_states (2n-1) : 27 (13 angle states + 14 voltage states)
457 [WLS] Measurements m : 68

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458 [WLS] Redundancy      : 68/27 = 2.519
459 [WLS] Convergence tol : 1e-06
460 [WLS] Max iterations  : 50
461
462
463 [WLS] ✓ Converged in 5 iterations (max|Δx| = 8.00e-11 < tol = 1e-06)
464 [WLS] Final objective J(x̂) = 0.446315
465 [WLS] Final ||r||      = 0.000668
466

```

Iteration summary:

Iter	r	rank(Gm)	cond(Gm)	max Δx
1	3.49788	27	65310	0.306458
2	0.238942	27	56670	0.0698145
3	0.0224277	27	61800	0.00164444
4	0.000668534	27	61900	2.98302e-06
5	0.000668068	27	61900	7.99916e-11

Estimated State vs CDF Load-Flow Ground Truth:

Bus	Name	V _est	V _true	Δ V	θ_est°	θ_true°	Δθ°
1		1.05999	1.06	-1e-05	0	0	0
2		1.045	1.045	-0	-4.98	-4.98	0
3		1.01	1.01	-0	-12.716	-12.72	0.004
4		1.01862	1.0186	2e-05	-10.322	-10.32	-0.002
5		1.02026	1.0203	-4e-05	-8.781	-8.78	-0.001
6		1.06999	1.07	-1e-05	-14.22	-14.22	0
7		1.06194	1.062	-6e-05	-13.366	-13.37	0.004
8		1.08999	1.09	-1e-05	-13.365	-13.37	0.005
9		1.05634	1.0563	4e-05	-14.944	-14.95	0.006
10		1.05132	1.0513	2e-05	-15.101	-15.1	-0.001
11		1.05708	1.0571	-2e-05	-14.792	-14.79	-0.002
12		1.05522	1.0552	2e-05	-15.074	-15.08	0.006

```

509 +-----+-----+-----+-----+-----+-----+-----+-----+
510 |    13 |         |    1.05044 |    1.0505 | -6e-05 |   -15.156 |    -15.16 |    0.004 |
511 +-----+-----+-----+-----+-----+-----+-----+-----+
512 |    14 |         |    1.03579 |    1.0358 | -1e-05 |   -16.036 |    -16.04 |    0.004 |
513 +-----+-----+-----+-----+-----+-----+-----+-----+

```

```

514
515 =====
516 SECTION 10 - BAD DATA ANALYSIS  [Pass 2]
517 =====

```

```

518 [BDD] m=68, n_states=27, dof=41
519 [BDD] Critical measurements ( $\Omega_{ii} \approx 0$ ,  $r_N$  forced to 0): 0
520 [BDD]  $r_N$  stats: min=0.0003 max=0.6506 mean=0.0552
521 [BDD] ✓ No bad data detected
522

```

```

523 [BDD] — LNR Test (threshold=3.0) — Top 10 —————

```

Rank	Idx	OrigIdx	Src	Type	Bus_i	Bus_j	z	$h(\hat{x})$	r	r_N	Flag
1	7	7	file	P_inj	1	—	2.32283	2.32327	-0.00044	0.651	OK
2	35	35	file	P_flow	1	2	1.56828	1.56792	0.00036	0.426	OK
3	6	6	file	P_inj	2	—	0.183	0.18312	-0.00012	0.224	OK
4	13	13	file	P_inj	5	—	-0.076	-0.07589	-0.00011	0.187	OK
5	45	45	file	P_flow	1	5	0.75552	0.75535	0.00017	0.182	OK
6	5	5	file	P_inj	3	—	-0.942	-0.94192	-8e-05	0.114	OK
7	46	46	file	P_flow	2	4	0.56139	0.56129	0.0001	0.109	OK
8	12	12	file	P_inj	4	—	-0.478	-0.47794	-6e-05	0.105	OK
9	44	44	file	P_flow	2	5	0.41513	0.41504	9e-05	0.097	OK
10	31	31	file	P_flow	5	6	0.44061	0.44053	8e-05	0.091	OK

```

547
548 [BDD] ✓ All normalised residuals  $\leq 3.0$  - no bad data
549

```

```

550 [LOOP] ✓ No bad data detected at pass 2.
551 [LOOP] Clean solution achieved.
552

```

```

553 =====
554 PIPELINE COMPLETE
555 =====

```

```

556 System observable (initial) : True
557 Stop reason                  : clean solution at pass 2
558 Measurements removed (bad)   : 1
559 Removed orig_idx set         : [68]

```

```
560      Bad data remaining (r_N>3.0) : 0
561  =====
562
563  Process finished with exit code 0
564
```